

TARGET-90

JKBOSE HUMANITIES CLASS- 11TH

BASIC MATHEMATICS

One Shot

PHYSICS

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Topics to be Covered

A

Fraction

B

Power

C

Angle

D

Trigonometry

* Maths is the language of universe



Topics to be Covered

E

Graph

F

Differentiation

*Imp***G**

Integration

*Imp***H**

..



Topic-1 Fraction

The fraction is a simple division/divide.

Type 1

$$\frac{1}{2} = 0.5$$

$$\frac{2}{3} = 0.\underline{66}$$

$$\frac{1}{4} = 0.25$$

$$\frac{4}{3} = 1.33$$

$$\frac{1}{3} = 0.33$$

$$\frac{1}{4} = 0.25$$





Fraction

The fraction is a simple division/divide.

Type 2

$$\begin{array}{r} 0.33 \\ \times 9 \\ \hline 3 \end{array}$$



Q $\underline{0.33} \times 9$

$$\frac{1}{3} \times 9 = 3$$

Q

$$\underline{0.75} \times \frac{4}{3}$$

$$\frac{3}{4} \times \frac{4}{3} = 1$$





Fraction

The fraction is a simple division/divide.

Type 3

$$\frac{3}{10} = 0.3$$

$$\frac{3}{100} = 0.03$$

$$\frac{49}{1000} = 0.049$$

$$3.84 \times 10 = 38.4$$

$$3.74 \times 10 = 37.4$$





Fraction

The fraction is a simple division/divide.

Type 4

$$678 \times 10^{-2} = \frac{678}{10^2} = \frac{678}{100} = 6.78$$





Topice-2 Power



When you multiply ~~one~~ same number:

$$2 \times 2 = 2^2$$

$$3 \times 3 = 3^2$$

$$4 \times 4 = 4^2$$

$$6 \times 6 \times 6 \times 6 = 6^4$$

$$x^2 \quad x^{-1/2}$$

$$x^{1/2}$$

$$x^{3/2}$$



$\times$



$=$





Power

$$x^{1/2} = \sqrt{x}$$

$$x^2 \times x^3 = x^5$$

Rules

1 ● $2^{1/2} = \sqrt{2}$

2 ● $2^{1/3} = \sqrt[3]{2}$

3 ● $x^0 = 1$

4 ● $10^0 = 1$

5 ● $3^2 \times 3^3 = 3^{2+3} = 3^5$





Power

$$\underline{2^2} \times \underline{2^3} = 2^{2+3} = 2^5$$

Rules

6 • $2^2 + 2^3 = 4 + 8 = 12$

7 • $5 \times 10^2 + 2 \times 10^2$

$$\underline{10^2} (5 + 2)$$

$$100 \times 7 = 700.$$





Power

$$\frac{1}{\text{Bohat sare log}} = 0$$

Rules

$$8 \rightarrow (2^3)^2 = 2^{3 \times 2} = 2^6$$

$$9 \rightarrow 54 \times 0 = 0$$

$$10 \rightarrow \frac{1}{\infty} = 0$$

$$\rightarrow \frac{1}{0} = \infty$$

$$\frac{1}{\infty} = 0$$

$$\frac{1}{0} = \infty$$





Power

Note karlo

$$\sqrt{4} = \sqrt{2^2} = 2$$

$$\sqrt{16} = \sqrt{4^2} = 4$$

$$\sqrt{2} = 1.4$$

$$\sqrt{3} = 1.7$$





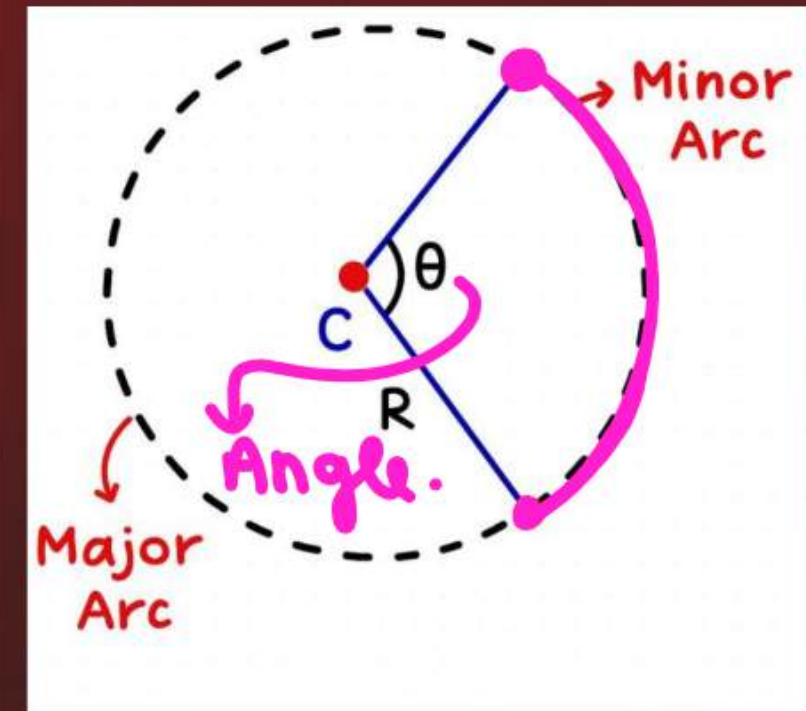
Angle

- Angle is the ratio of Arc and radius.

$$\theta = \frac{\text{arc}}{\text{radius}} = \frac{\cancel{2\pi}}{\cancel{2\pi}} = 1$$

- S.I. unit of angle = "Radian"

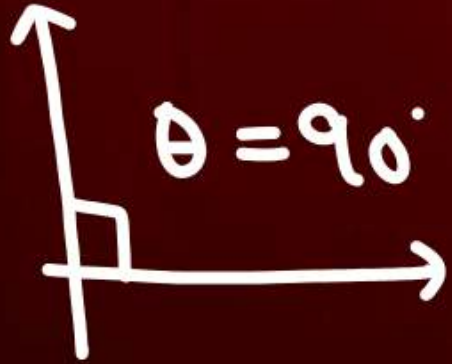
- Other unit of angle = "Degree"



$$\theta = 0^\circ$$



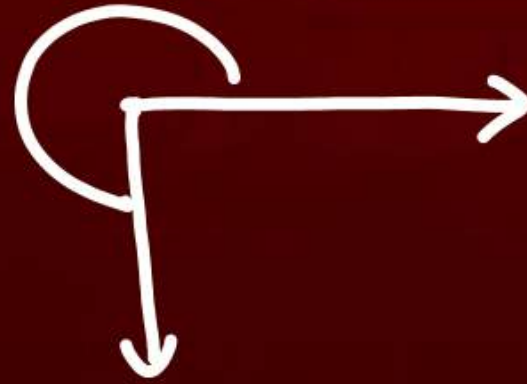
$$\theta = 90^\circ$$



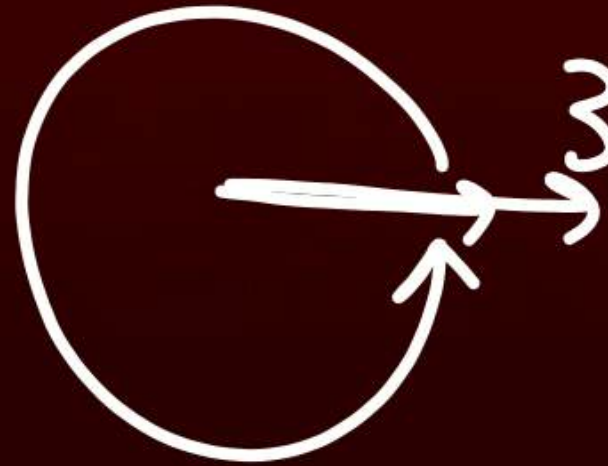
$$180^\circ = \pi \text{ rad}$$



$$270^\circ$$



$$360^\circ$$



$$180^\circ = \pi \text{ rad}$$

$$180^\circ = 3 \text{ rad}$$

- $180^\circ = \pi \text{ rad}$

$$\Rightarrow 180 \times 1^\circ = \pi \text{ rad}$$

$$1^\circ = \frac{\pi}{180} \text{ rad}$$

- $180^\circ = \pi \text{ rad}$

$$180^\circ = \pi \times 1 \text{ rad}$$

$$\frac{180^\circ}{\pi} = 1 \text{ rad}$$



Angle

$$180^\circ = \pi \text{ rad}$$

$$1^\circ = \frac{\pi}{180} \text{ rad}$$

$$1 \text{ rad} = \frac{180^\circ}{\pi}$$





Angle



$$30^\circ = \frac{\pi}{6} \text{ rad}$$

$$30 \times 1^\circ = 30 \times \frac{\pi}{180} \text{ rad} \\ = \frac{\pi}{6} \text{ rad}$$



$$\frac{\pi}{6} \text{ rad} = \text{---}$$

$$\frac{\pi}{6} \text{ rad} = \frac{\pi}{6} \times 1 \text{ rad} \\ \frac{\pi}{6} \times \frac{180^\circ}{\pi} = 30^\circ$$



$$60^\circ = \frac{\pi}{3} \text{ rad}$$

$$60^\circ = 60 \times 1^\circ \\ = 60 \times \frac{\pi}{180} \text{ rad} \\ = \frac{\pi}{3} \text{ rad}$$



$$\frac{\pi}{3} \text{ rad} = \text{---}$$

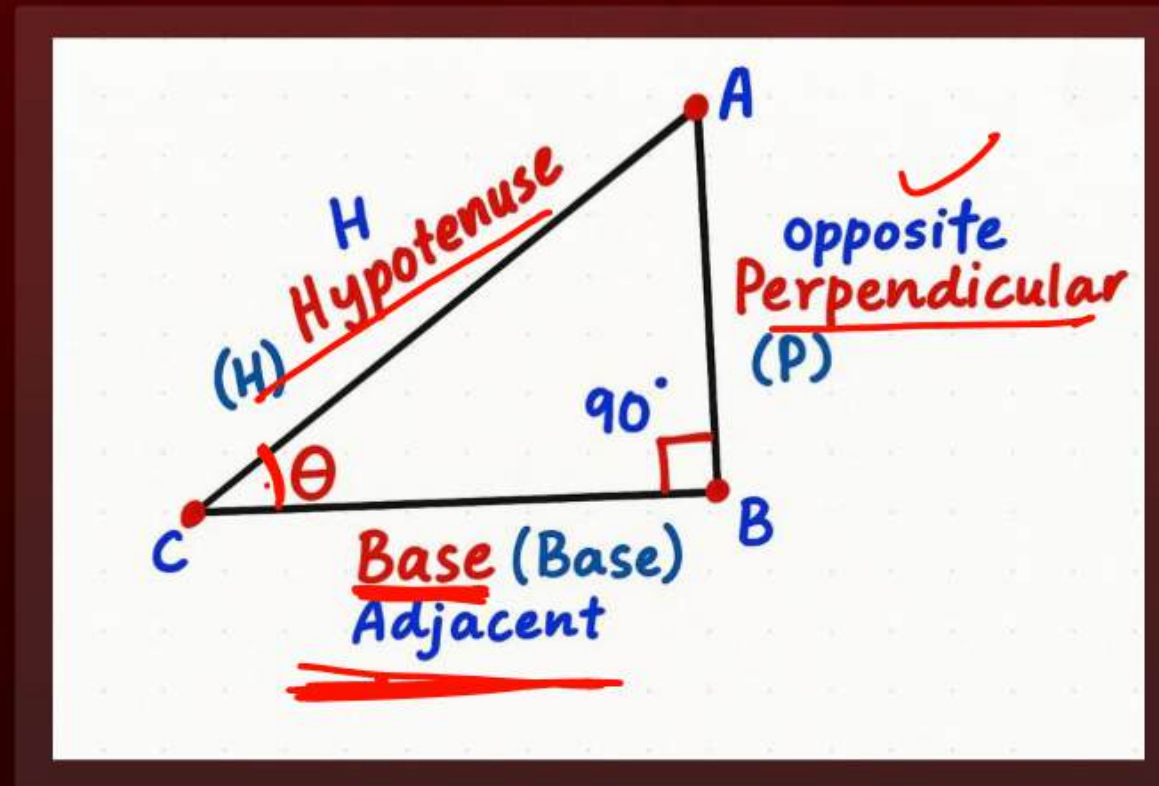
$$\frac{\pi}{3} \text{ rad} = \frac{\pi}{3} \times 1 \text{ rad} \\ \frac{\pi}{3} \times \frac{180^\circ}{\pi} = 60^\circ$$



Topic-4 Trigonometry

* class 10th

- The study of side and Angle of Right triangle

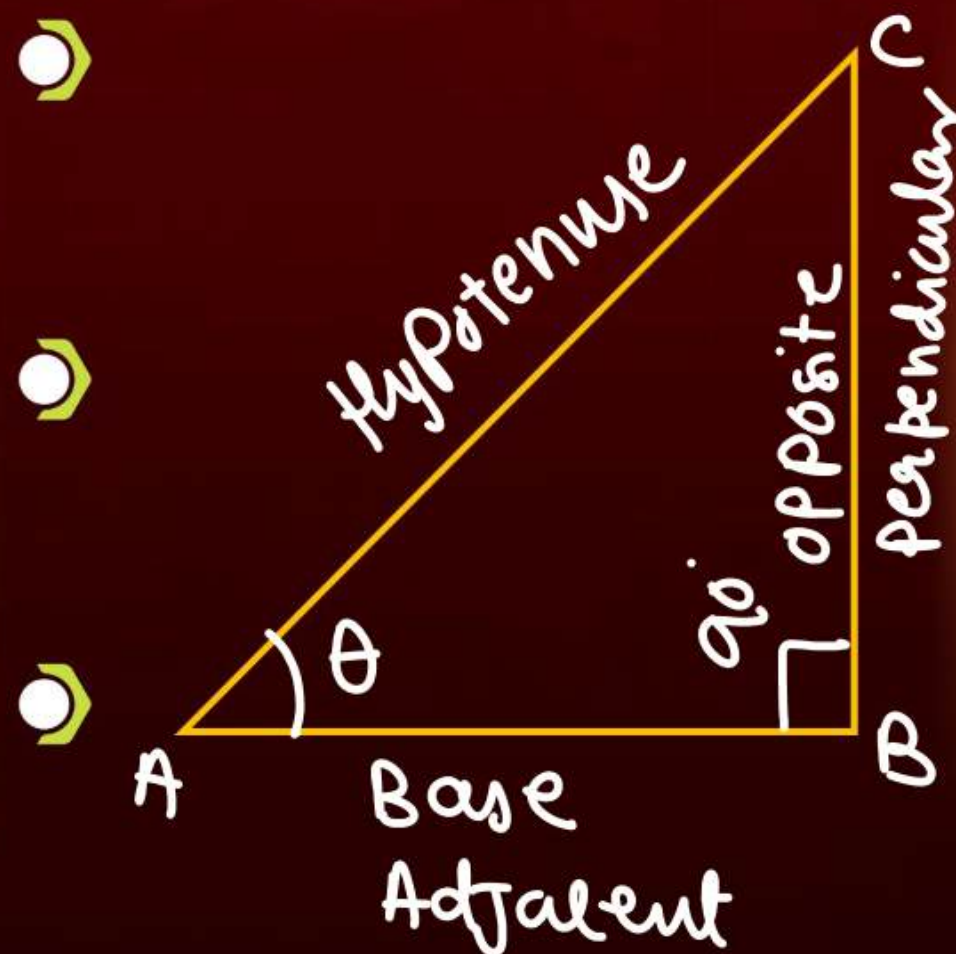




Trigonometry

Trigonometric Ratio

- $\sin\theta = \frac{P}{H} = \frac{O}{I}$
- $\cos\theta = \frac{B}{H} = \frac{A}{I}$
- $\tan\theta = \frac{P}{B} = \frac{O}{A}$
- $\operatorname{cosec}\theta = \frac{H}{P} = \frac{I}{O}$
- $\sec\theta = \frac{H}{B} = \frac{I}{A}$
- $\cot\theta = \frac{B}{P} = \frac{A}{O}$



- $\sin\theta = \frac{1}{\operatorname{Cosec}\theta}$
- $\cos\theta = \frac{1}{\sec\theta}$
- $\tan\theta = \frac{1}{\cot\theta}$



Trigonometry

Trick

$$\sin \theta = \frac{P}{H} \quad \cos = \frac{B}{H} \quad \tan \theta = \frac{P}{B}$$

Sin

$$\frac{P_{\text{andit}}}{H_{\text{ar}}}$$

Cos

$$\frac{B_{\text{adri}}}{H_{\text{ar}}}$$

Tan

$$\frac{P_{\text{rasad}}}{B_{\text{ole}}}$$



• $\tan\theta = \frac{\sin\theta}{\cos\theta}$

$$\tan\theta = \frac{\text{P}}{\text{H}} \quad \text{Sin}\theta$$

$$\frac{\text{B}}{\text{H}} \quad \text{Cos}\theta$$

$$\tan\theta = \frac{\sin\theta}{\cos\theta}$$



Trigonometry Ratios



Trigonometry Table :-

$$\tan \theta = \frac{\sin \theta}{\cos \theta}$$

Degrees	0°	30°	45°	60°	90°
Radians	0	$\pi/6$	$\pi/4$	$\pi/3$	$\pi/2$
$\sin \theta$	0	$\frac{1}{2}$	$\frac{1}{\sqrt{2}}$	$\frac{\sqrt{3}}{2}$	1
$\cos \theta$	1	$\frac{\sqrt{3}}{2}$	$\frac{1}{\sqrt{2}}$	$\frac{1}{2}$	0
$\tan \theta$	0	$\frac{1}{\sqrt{3}}$	1	$\sqrt{3}$	$\frac{1}{0} = \infty$

→ not define.

← Arranged Reverse $\sin \theta$ value →



Trigonometry



θ	0°	30°	45°	60°	90°	120°	135°	150°	180°
$\sin \theta$	0	$\frac{1}{2}$	$\frac{1}{\sqrt{2}}$	$\frac{\sqrt{3}}{2}$	1	$\frac{\sqrt{3}}{2}$	$\frac{1}{\sqrt{2}}$	$\frac{1}{2}$	0
$\cos \theta$	1	$\frac{\sqrt{3}}{2}$	$\frac{1}{\sqrt{2}}$	$\frac{1}{2}$	0	$-\frac{1}{2}$	$-\frac{1}{\sqrt{2}}$	$-\frac{\sqrt{3}}{2}$	-1
$\tan \theta$	0	$\frac{1}{\sqrt{3}}$	1	$\sqrt{3}$	∞	$-\sqrt{3}$	-1	$-\frac{1}{\sqrt{3}}$	0



Trigonometry

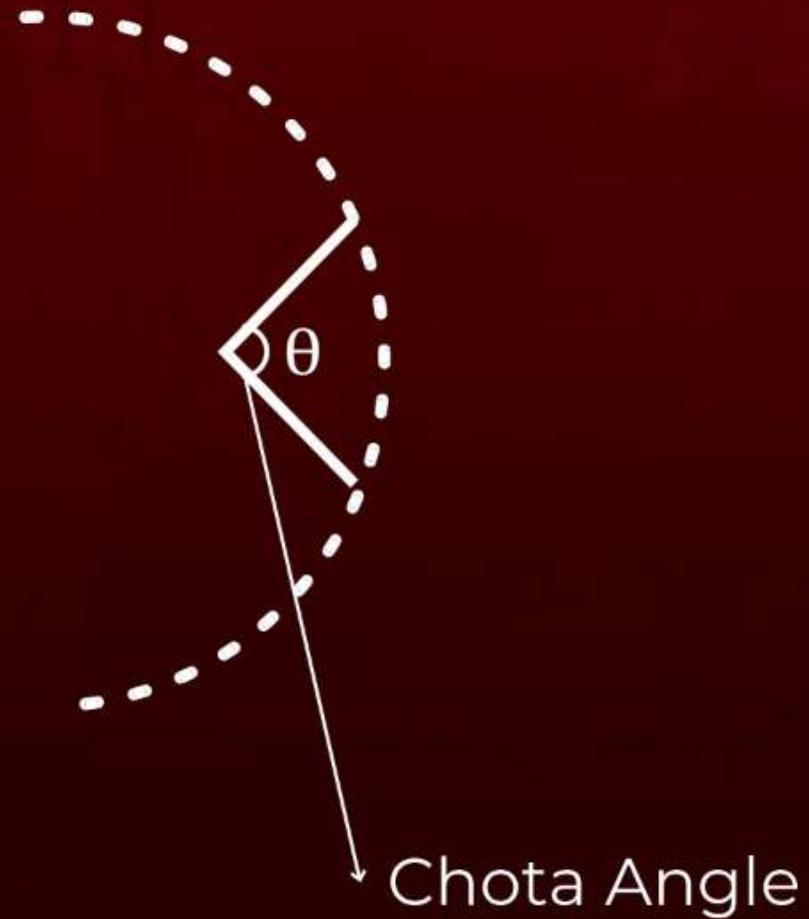
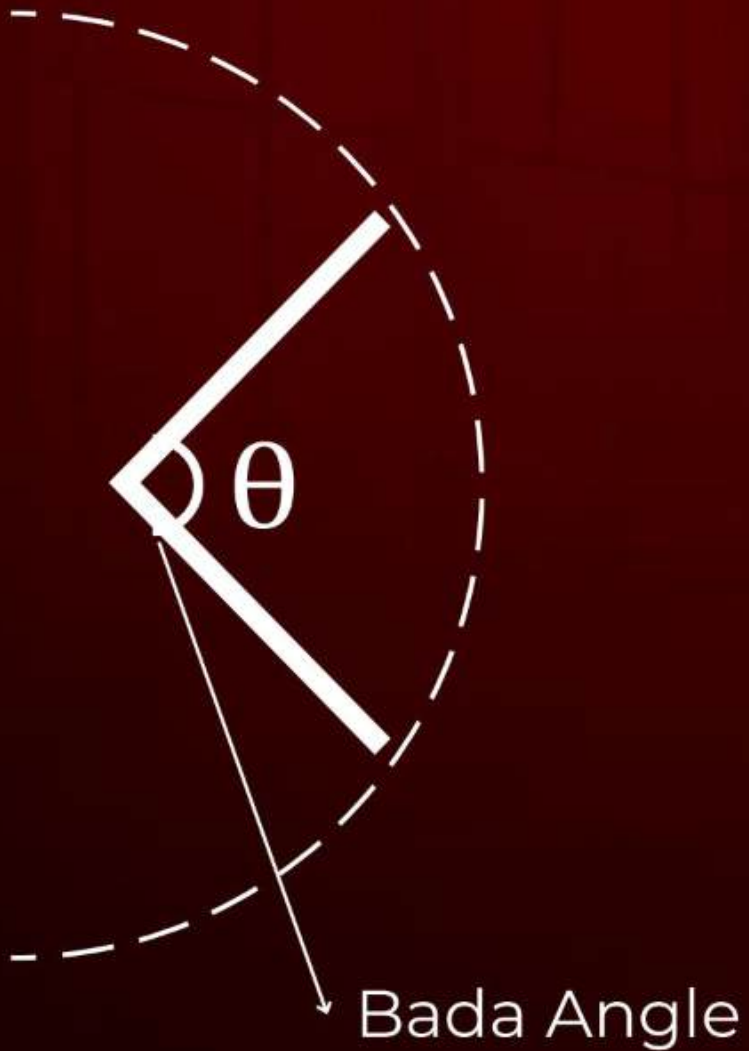
Small angle approximation

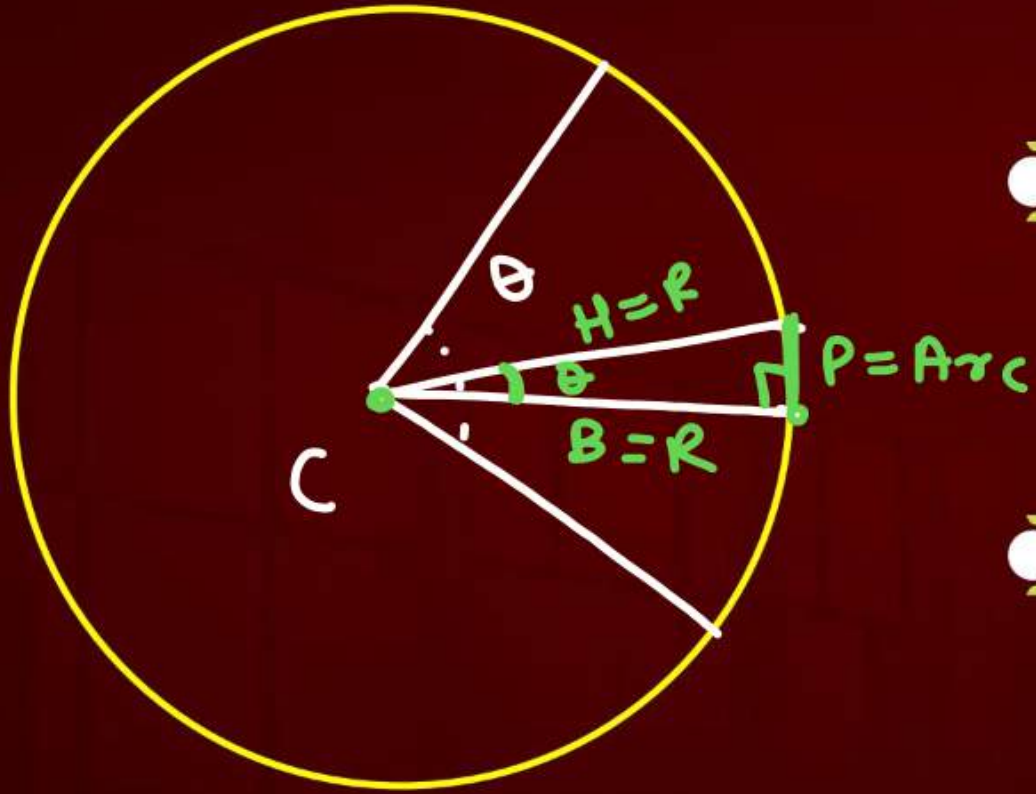
Bada Angle Ko Chota Kardiya

$$\underline{\sin\theta} = \theta$$

$$\underline{\tan\theta} = \theta$$

$$\underline{\cos\theta} = 1$$





$$\bullet \sin \theta = \frac{P}{H} = \frac{\text{Arc } C}{R} = \theta$$

$$\bullet \tan \theta = \frac{P}{B} = \frac{\text{Arc } C}{R} = 0$$

$$\bullet \cos \theta = \frac{B}{H} = \frac{R}{R} = 1$$



Topic-5 Graph ki kahani



Equation

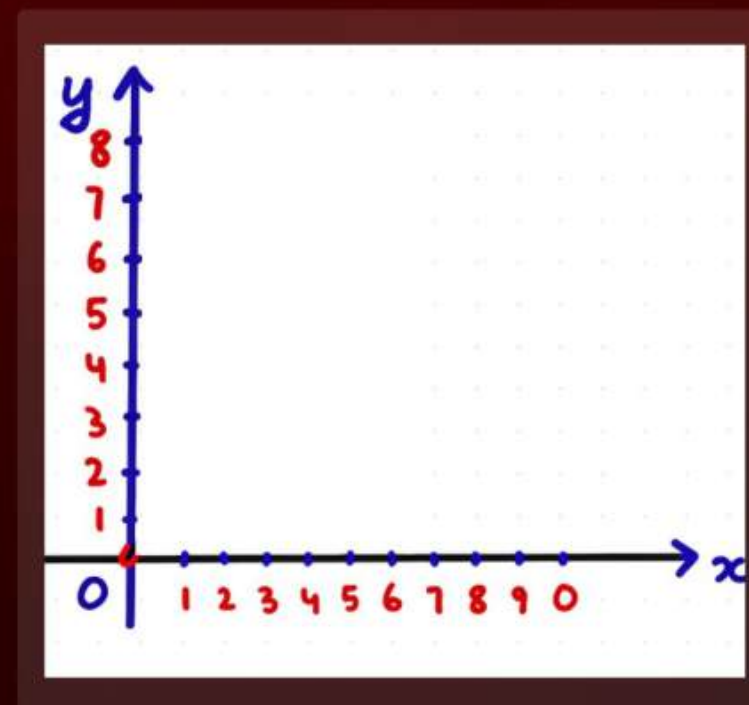
When you write English statement in math, form having Const (1, 2, 3, 4, 5 etc) (+, -, x, /) and variable (x, y).

(a) Straight Line

Cost of pen is twice the cost of pencil.

$$y = 2x$$

x	1	2	3	4	5
y	2	4	6	8	10



• $x = 1$
 $y = 2x$

$y = 2 \times 1$
 $y = 2$

• $x = 2$
 $y = 2 \times x$

$y = 2 \times 2$
 $y = 4$

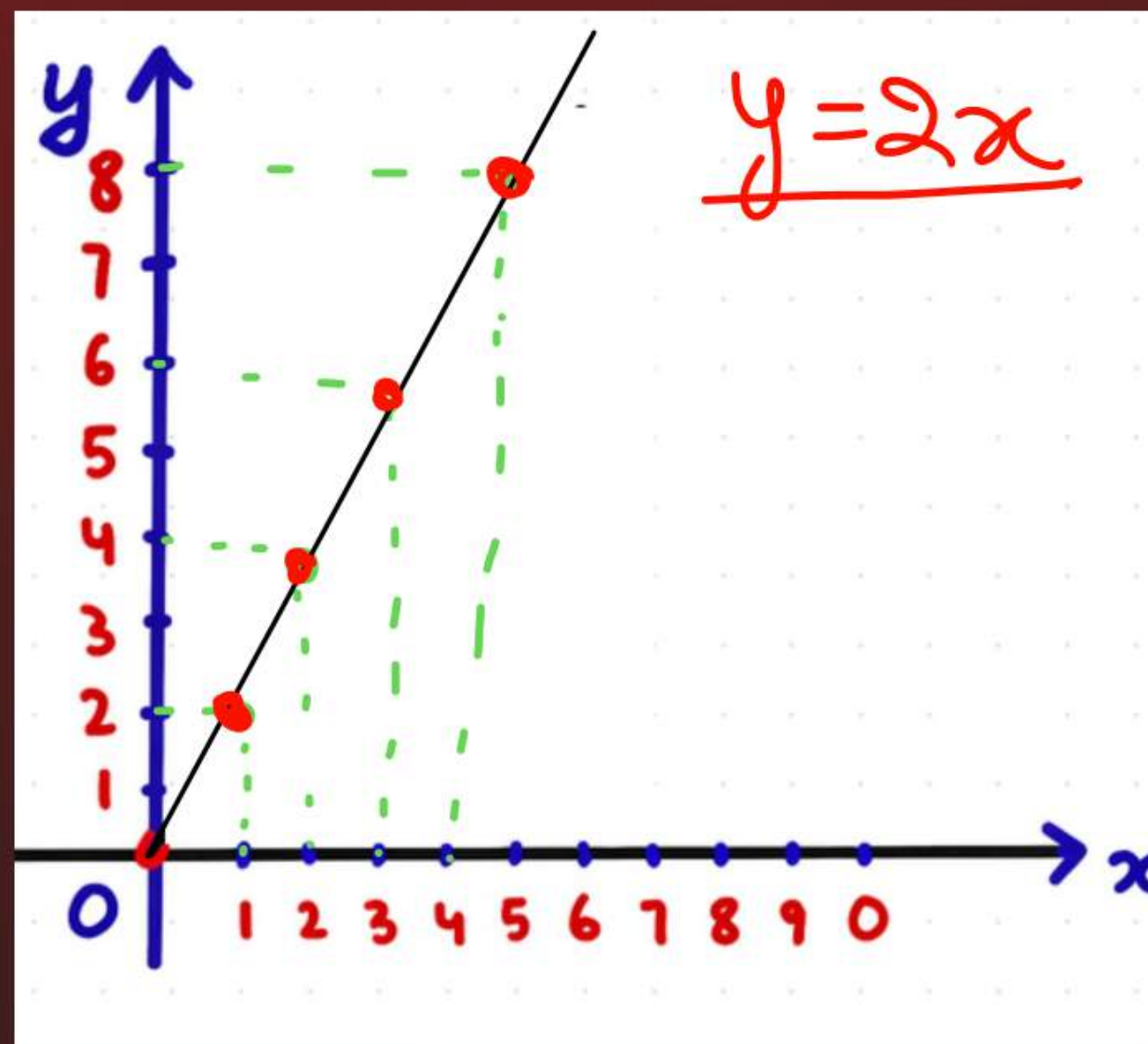
• $x = 3$
 $y = 2x$

$y = 2 \times 3$
 $y = 6$

• $x = 4$
 $y = 2x$

$y = 2 \times 4$
 $y = 8$

x	1	2	3	4	5
y	2	4	6	8	10



- $y = 2x + 0$

Equation Straight line equation

$$y = mx + c$$

Diagram illustrating the components of the straight line equation $y = mx + c$:

- y is labeled as the y-axis.
- m is labeled as the Slope.
- x is labeled as the x-axis.
- c is labeled as the intercept.



Graph ki kahani

$$y = 2x + 2$$

$$x = 0$$

$$y = 2 \times 0 + 2$$
$$y = 0 + 2 = 2$$

$$x = 1$$

$$y = 2 \times 1 + 2$$
$$y = 2 + 2 = 4$$

$$x = 2$$

$$y = 2 \times 2 + 2$$
$$y = 4 + 2 = 6$$

x	0	1	2
y	2	4	6

x	0	1	2
y	2	4	6

$y = 2x + 2$

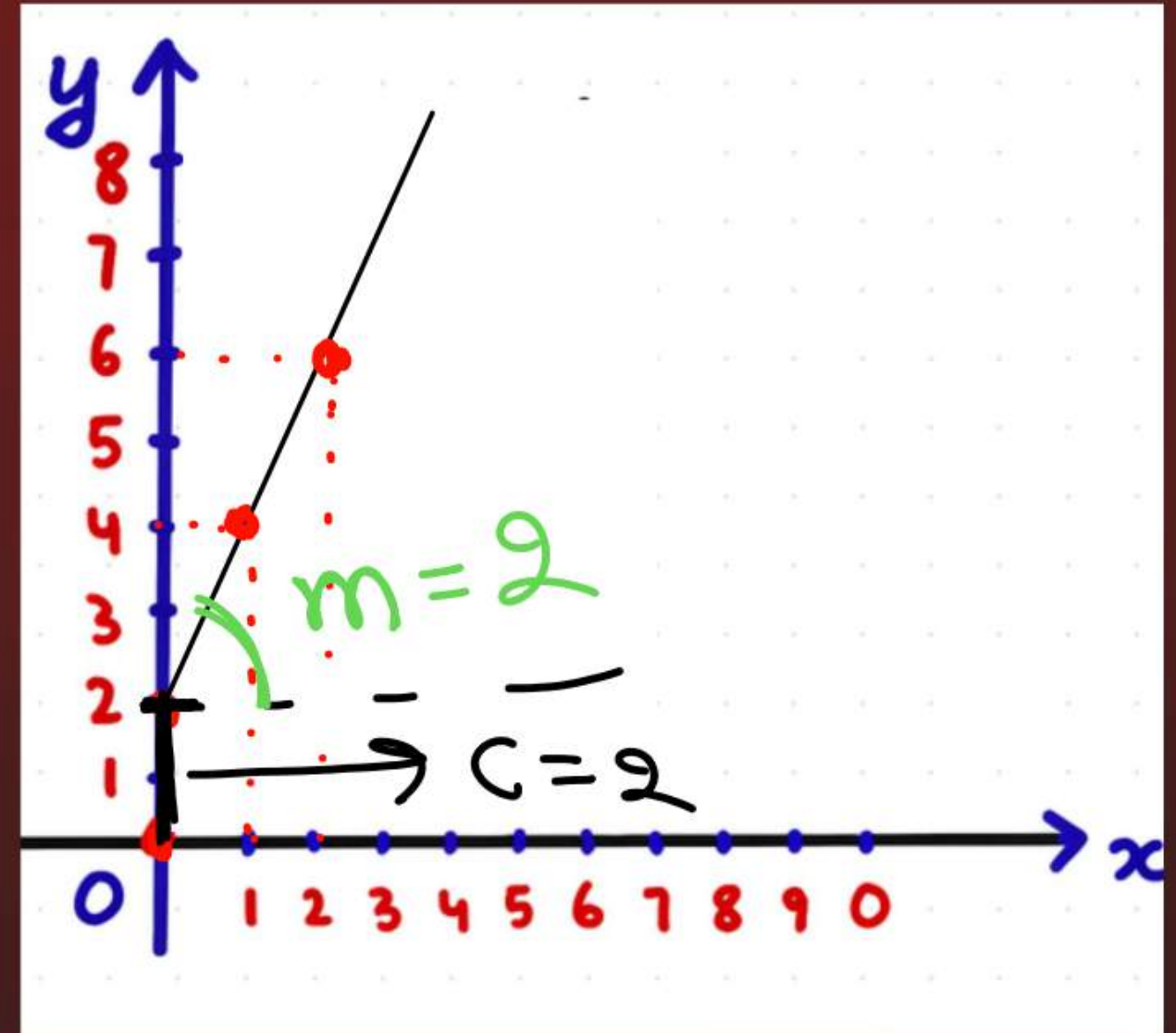
y-axis Slope x axis

intercept

$$y = 3x + 2$$

$$y = mx + c$$

$m = 3$ $c = 2$





Graph ki kahani

$$y = 2x + 2$$

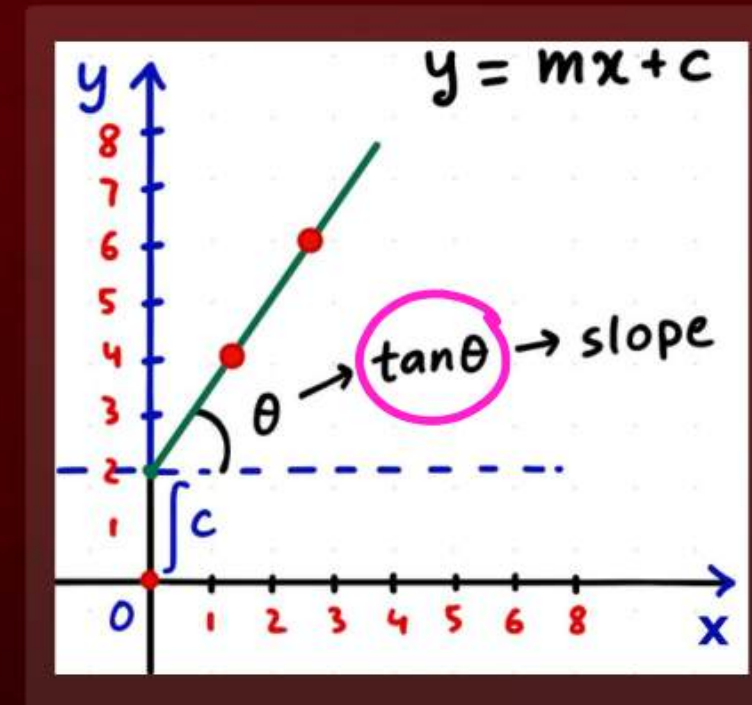
$$y = mx + c$$

y-axis

x-axis

Slope

y-intercept





Question

$$y = 3x + 5$$

$$y = 3x + 5$$

$$\checkmark m = ?$$

$$\checkmark c = ?$$

$$y = \underline{3}x + 5$$

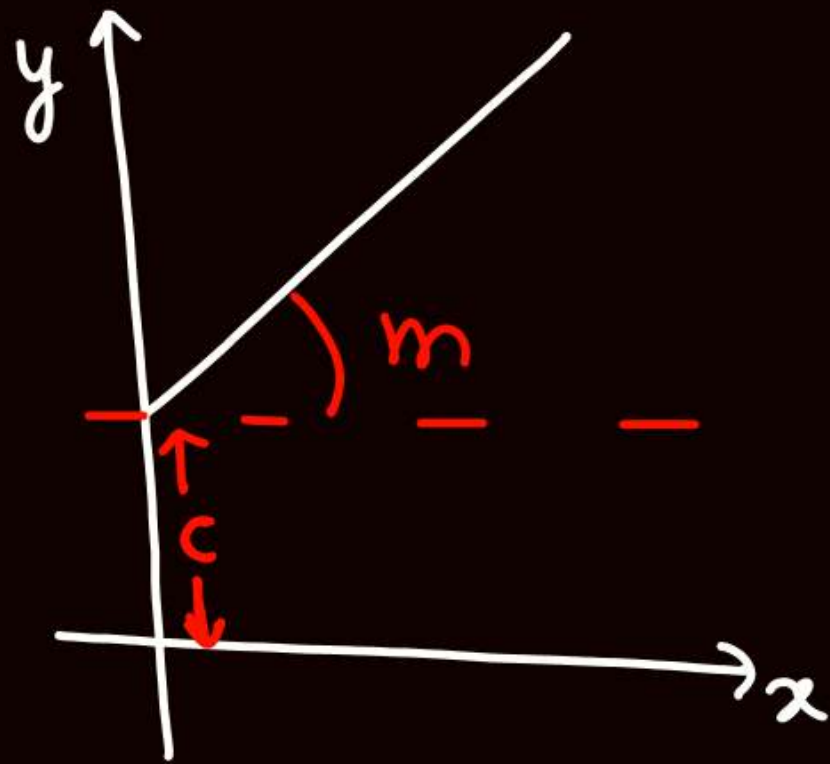
$$m = 3$$

$$c = 5$$

1. (*) $y = mx + c$

$$y = 5x + 3$$

$$y = 3x$$

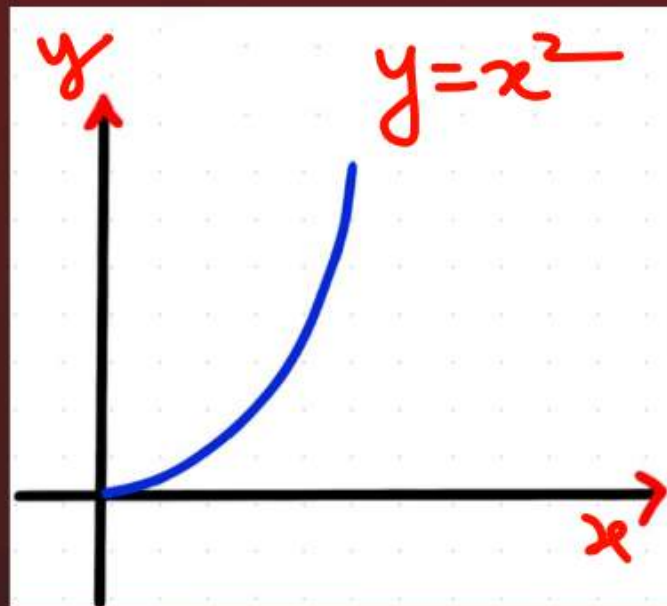


$$y = 2x$$

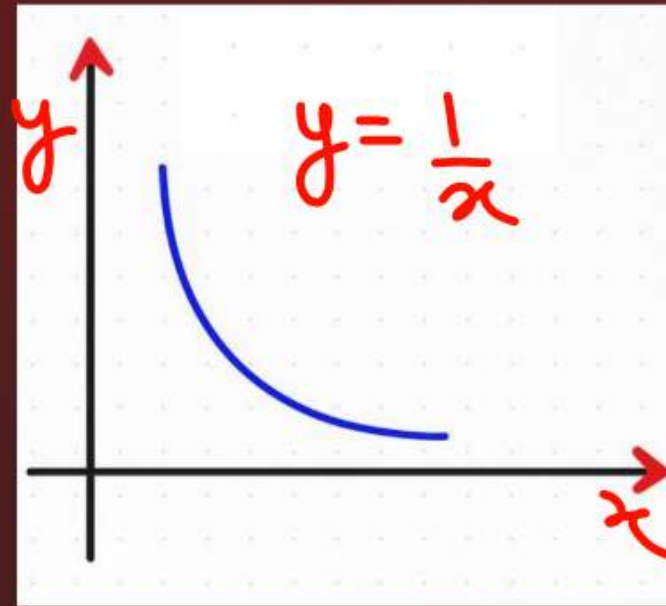


Graph ki kahani

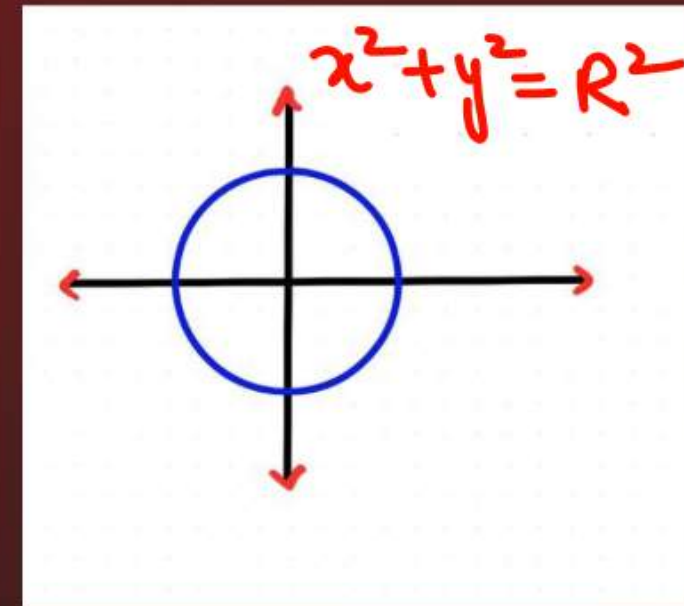
Types of Graphs



Parabolic Graph



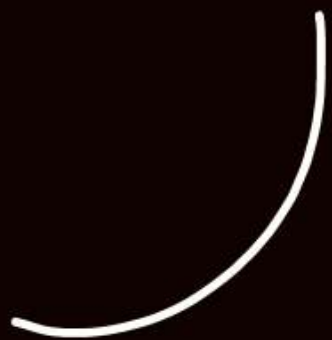
Hyperbolic Graph



Circle Graph



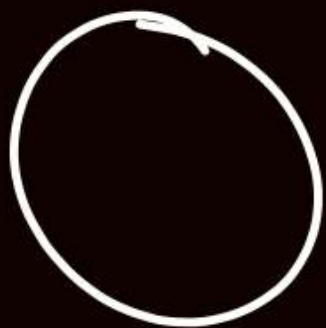
$$y = mx + c$$



$$y = x^2$$



$$y = \frac{1}{x}$$



$$x^2 + y^2 = R^2$$

Calculus

(Newton)

+
-
×
÷

Maths

(i)

Differentiation

$$\frac{dy}{dx}$$

Rules

(ii)

Integration

$$\int y dx$$

Rules



Differentiation

Bada change $\rightarrow \frac{\Delta y}{\Delta x}$

$\frac{dy}{dx}$
chota change

✓ Cost of pen is equal to twice the cost of pencil.

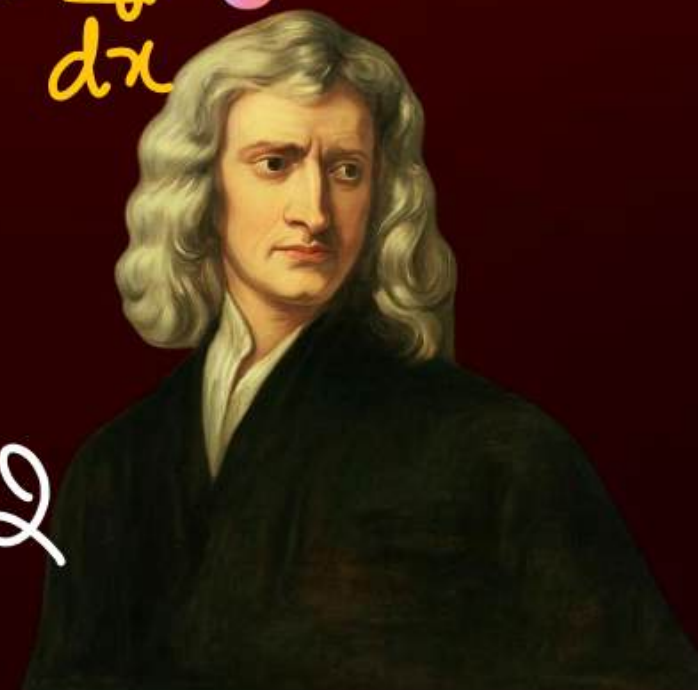
$$y = 2x$$

	$x_1 \quad x_2$					
x	0	1	2	3	4	5
y	0	2	4	6	8	10
	$y_1 \quad y_2$					

Differentiation = $\frac{\text{Small change in } y}{\text{Small change in } x} = \frac{y_2 - y_1}{x_2 - x_1} = \frac{\Delta y}{\Delta x} = \frac{dy}{dx}$

Wajahid sir, tumne acha padhaya!

differentiation $\frac{y_2 - y_1}{x_2 - x_1} = \frac{\Delta y}{\Delta x} = \frac{dy}{dx} = \frac{y_2 - y_1}{x_2 - x_1} = \frac{2 - 0}{1 - 0} = \frac{2}{1} = 2$





Differentiation

“Newton give the the concept of the differentiation.”

$\frac{dy}{dx}$ = means Torna ✓



Newton mera dost
hai re baba...





Differentiation

Rules of Differentiation

1. $y = c$
 $\frac{dy}{dx} = \frac{dc}{dx} = 0$

2. $y = x^n$
 $\frac{dy}{dx} = nx^{n-1}$

3. $y = ca$
 $\frac{dy}{dx} = \frac{dca}{dx} = c \frac{da}{dx}$

4. $\frac{d}{dx} \sin x = \cos x$

5. $\frac{d}{dx} \cos x = -\sin x$

6. $\frac{d}{dx} \tan x = \sec^2 x$



Differentiation

$\csc x$

$\csc \theta$

Rules of Differentiation

$$7 \Rightarrow \frac{d}{dx} \underline{\csc x} = -\csc x \cot x$$

$$8 \Rightarrow \frac{d}{dx} \sec x = \sec x \cdot \tan x$$

$$9 \Rightarrow \frac{d}{dx} \underline{\cot x} = -\csc^2 x$$

$$10 \Rightarrow \frac{d e^x}{dx} = e^x$$

$$11 \Rightarrow \frac{d}{dx} \log x = \frac{1}{x}$$



Differentiation

$$x^n = n x^{n-1}$$

Rules of Differentiation

Imp 5 mark

12 Inside outside rule
 $y = \sin x^2$

$$\frac{dy}{dx} = 2x^1 \cos x^2$$
$$= \underline{2x} \cos x^2$$

$$\text{Q } \frac{dy}{dx} = \frac{d}{dx} \sin(x^3)$$
$$= 3x^2 \cos x^3$$

$$\text{Q } y = \sin(x^2)$$
$$\frac{dy}{dx} = \frac{d}{dx} \sin(x^2)$$
$$= \underline{2x \cos x^2}$$



Differentiation

Rules of Differentiation

13



Sum Rule



$$y = x^3 + x^2 + \sin x + 4$$

$$\frac{dy}{dx} = \frac{d}{dx} (x^3 + x^2 + \sin x + 4)$$

$$= \frac{d}{dx} x^3 + \frac{d}{dx} x^2 + \frac{d \sin x}{dx} + \frac{d 4}{dx}$$

$$= 3x^2 + 2x^1 + \cos x + 0$$

$$= 3x^2 + 2x + \cos x$$



Differentiation

$$x^3 = 3x^{3-1}$$

$$= \underline{3x^2}$$

Rules of Differentiation



Multiplication Rule

$$y = u \cdot v$$

$$\frac{dy}{dx} = \frac{d(uv)}{dx} = u \frac{dv}{dx} + v \frac{du}{dx}$$

Q $y = \frac{x^3}{u} \cdot \frac{\sin x}{v}$

$$\begin{aligned} \frac{dy}{dx} &= \frac{d}{dx} \overset{u}{x^3} \cdot \overset{v}{\sin x} = x^3 \frac{d \sin x}{dx} + \sin x \frac{dx^3}{dx} \\ &= x^3 \cos x + \sin x \cdot 3x^2 \end{aligned}$$



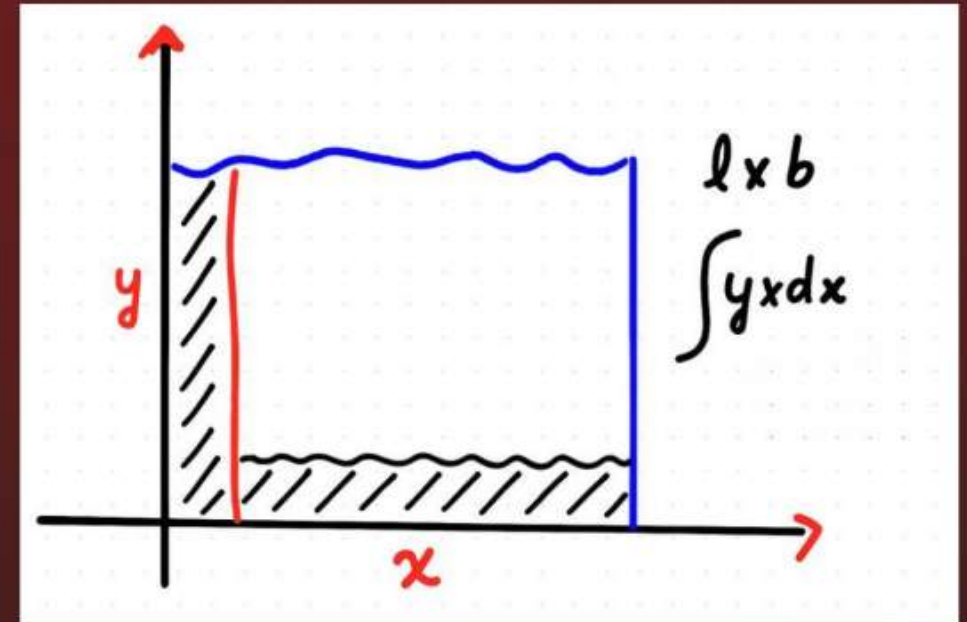
Integration

Integration dene walla Bi Newton Chacha hai ✓

$\int y dx$ = means Jorna



Newton mera dost
hai re baba...





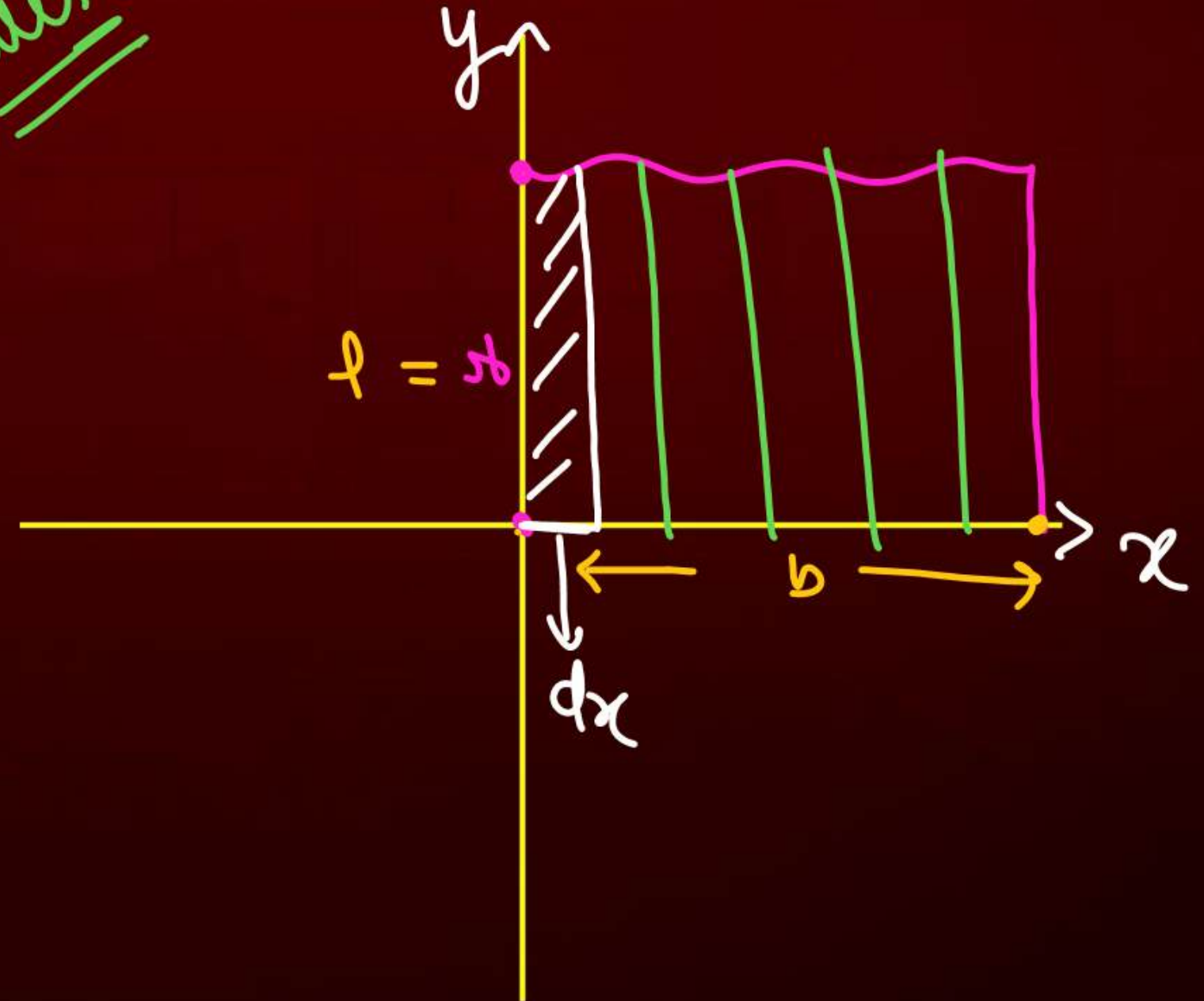
Graphical Explanation of integration

• Area = $l \times b$

• Small area = $y dx$

$$\int \text{small area} = \int y dx$$

Rules





Graphical Explanation of integration

• $\int y dx$ (Integration)



Integration

Rules of Integration

$$\textcircled{1} \Rightarrow \int x^n dx = \frac{x^{n+1}}{n+1}$$

$$\textcircled{3} \Rightarrow \int e^x dx = e^x$$

$$\textcircled{2} \Rightarrow \int \sin x \cdot dx = -\cos x$$

$$\textcircled{4} \Rightarrow \int \frac{1}{x} dx = \log x$$

$\ln x$

Integration

✓
 $\int y dx$
 without limit

$\int_{x_1}^{x_2} y dx$
 limit



Question

A

$$y = x^3$$

$$\int y dx \quad \int x^n dx = \frac{x^{n+1}}{n+1}$$
$$\int x^3 dx = \frac{x^{3+1}}{3+1}$$
$$= \frac{x^4}{4} + C$$

B

$$y = x^3$$

$$\int_{x_1=1}^{x_2=2} y dx = \int_1^2 x^3 dx$$
$$= \left[\frac{x^{3+1}}{3+1} \right]_1^2 = \left[\frac{x^4}{4} \right]_1^2$$

$$= \left[\frac{2^4}{4} - \frac{1^4}{4} \right]$$

$$= \left[\frac{2^4}{4} - \frac{1^4}{4} \right]$$

$$= \frac{16}{4} - \frac{1}{4}$$

$$= \frac{16-1}{4} = \frac{15}{4}$$



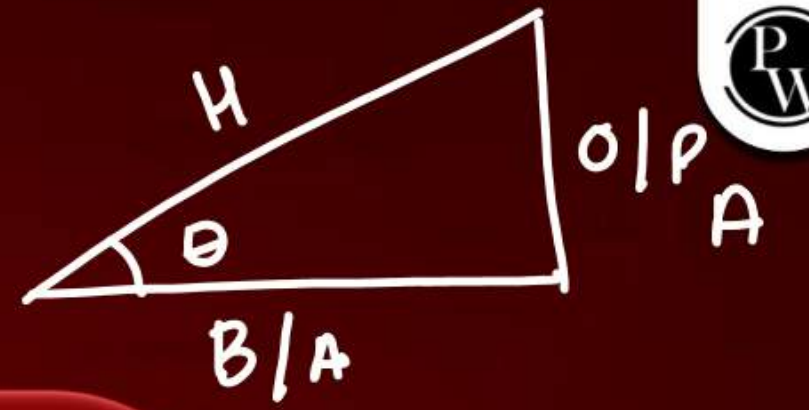
Homework



Revise



notes



**Thank
You**